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Toda extensión sobre un entero es un módulo finitamente generado

Teorema 1. *Sea $A \subset B$ una extensión de anillos y $b \in B$ un elemento entero sobre A*

$$\implies A[b] \text{ es un } A\text{-módulo finitamente generado.}$$

***Demostración*:** Como b es entero sobre A, existe un polinomio mónico $p \in A[x]$ dado por $p(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0$ tal que $p(b) = 0$. Por tanto,

$$b^n = -a_{n-1}b^{n-1} - \cdots - a_0 \implies \forall m \geqslant n : b^n b^{m-n} = -a_{n-1}b^{n-1}b^{m-n} - \cdots - a_0 b^{m-n}.$$

Entonces, $1, b, b^2, \ldots, b^{n-1}$ generan $A[b]$ como A-módulo. ■

Referenciado en

- `cor-extension-entera-anillos-transitiva`

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